

CIRCULATION

Chapter

12

Major Concept

In this Unit you will learn:

- Circulation
- Blood Circulatory System of Man
- Heart
- Blood Vessels
- Blood Pressure and its Measurement
- Cardiovascular Disorders
- Lymphatic System of Man





CIRCULATION

Diffusion alone is not adequate for transporting chemicals over macroscopic distances in animals for examples to transport glucose from the digestive tract and oxygen from the lungs to the brain of animal. The time it takes for a substance to diffuse from one place to another is proportional to the square of the distance the chemical will travel. The **circulatory system** solves this problem by ensuring that no substance must diffuse very far to enter or leave a cell. By transporting fluid throughout the body, it functionally connects the aqueous environment of the body cells to the organs that exchange gases, nutrients and dispose off wastes. In the lungs of a mammal, for example, oxygen from inhaled air diffuses across a thin epithelium and into the blood, while carbon dioxide diffuses in the opposite direction. The circulatory system then carries the oxygen rich blood to all parts of the body. As the blood streams through the tissues within microscopic vessels called capillaries, chemicals are transported between the blood and the interstitial fluid that directly bathes the cells.

12.1 TYPES OF CIRCULATORY SYSTEMS

The blood circulatory system is basically of two types in animals open type and closed type.

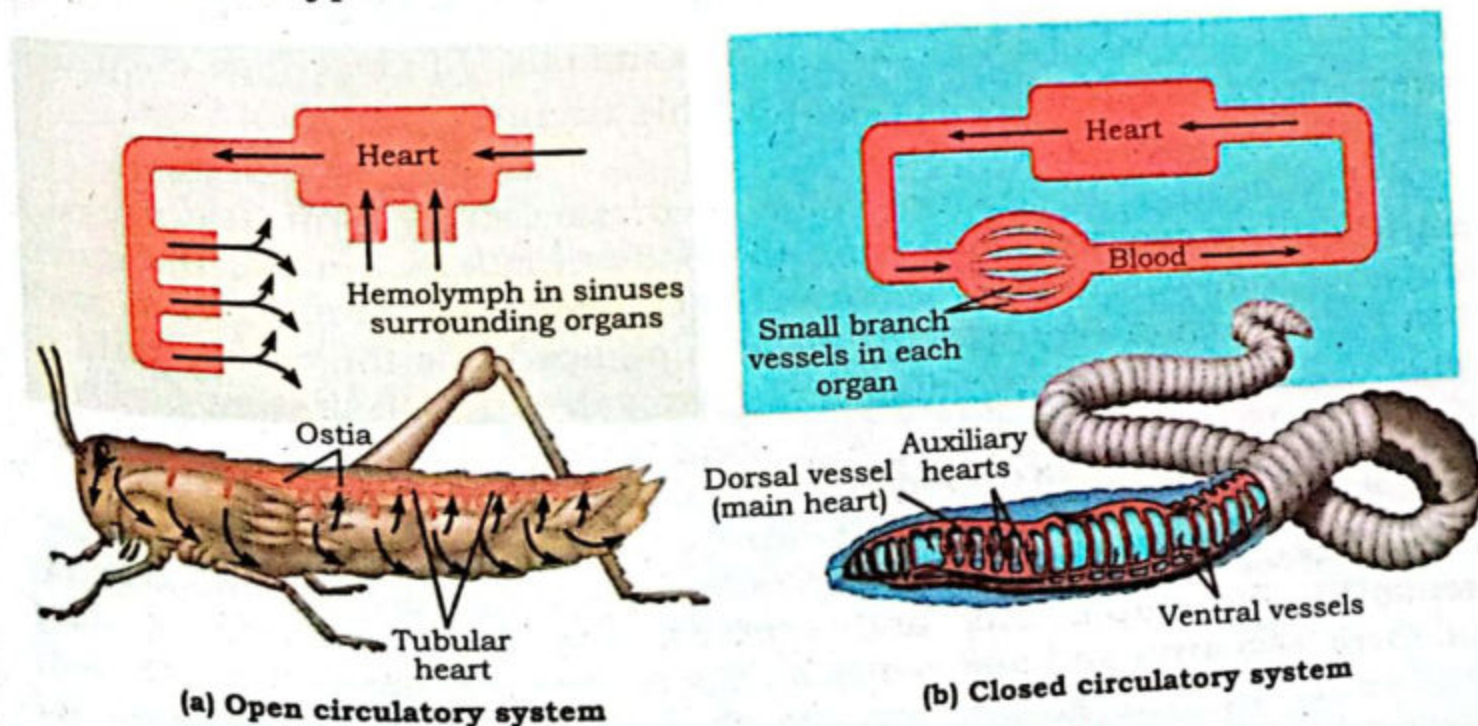


Fig: 12.1 Open and close type circulation

A. Open circulatory system

A gastrovascular cavity is inadequate for internal transport within animals having many layers of cells, especially in terrestrial animals. In arthropods and most molluscs, blood bathes the internal organs directly, this arrangement of called an **open circulatory system**.



There is no distinction between blood and interstitial fluid, so the general body fluid is more correctly termed as **haemolymph**. One or more hearts pump the haemolymph into an interconnected system of sinuses, which are spaces surrounding the organs. Here chemical exchange occurs between the haemolymph and body cells. In grasshopper and other arthropods, heart is an elongated tube located dorsally, when the heart contracts it pumps haemolymph into dorsal aorta which empties into the **haemocoel**. When the heart contracts openings called ostia are closed when the heart relaxes, the haemolymph is sucked back into the heart by way of the **ostia**. The haemolymph of grasshopper and other insects is colourless because it does not contain **haemoglobin** or any other respiratory pigment.

B. Closed type circulatory system

In a closed circulatory system, blood is confined to vessels and is distinct from the interstitial fluid. One or more heart pumps blood into large vessels that branch into smaller ones coursing through the organs. Here materials, are exchanged between the blood and the interstitial fluid bathing the cell. Annelids, some molluscs (octopus and squid) and vertebrates have closed type circulatory system.

During the course of vertebrate evolution, the heart has become increasingly complex, starting with the two chambered heart of fish and culminating in the four chambered heart of birds and mammals. In close type circulation system, circulation is either single or double circuit.

i. Single circuit circulation

The heart of fish is two chambered consisting of an atrium and a ventricle, deoxygenated blood from the body circulation enters the atrium through a thin-walled sinus venosus which open into a muscular ventricle upon contraction of ventricle, the blood is pumped into the ventral aorta via conus arteriosus. All these chambers have valves which prevent backward flow of blood.

ii. Double circuit circulation

From amphibians onwards up to the mammals, the circulatory system has double circuit circulation with separate pulmonary and systemic circulation. In amphibians and reptiles, the heart consists of three chambers, two atria and one ventricle, the oxygenated blood from the lungs returned to the left atrium through the pulmonary veins, whereas the deoxygenated blood from the body is passed to the right atrium via sinus venosus by the anterior and posterior vena cavae. These two types of blood streams remain separated due to atrial septum in between the right and left atria, but get mixed to some extent within the ventricle. Since the oxygenated and deoxygenated blood are mixed, the circulation is known as **incomplete double circulation**.



In crocodiles, birds and mammals the heart is four chambered with two atria and two ventricles. The right atrium and right ventricle are completely separated from the left atrium and left ventricle by inter-atrial and inter-ventricular septa. The right side receives deoxygenated blood and the left side oxygenated blood. The blood circulates through the heart twice. Once as deoxygenated blood on the right side, from where it is pumped to the lungs for oxygenation (pulmonary circulation). Next time as oxygenated blood on the left side to be distributed to all the parts of body except lungs (systemic circulation). Thus it is known as **complete double circuit**

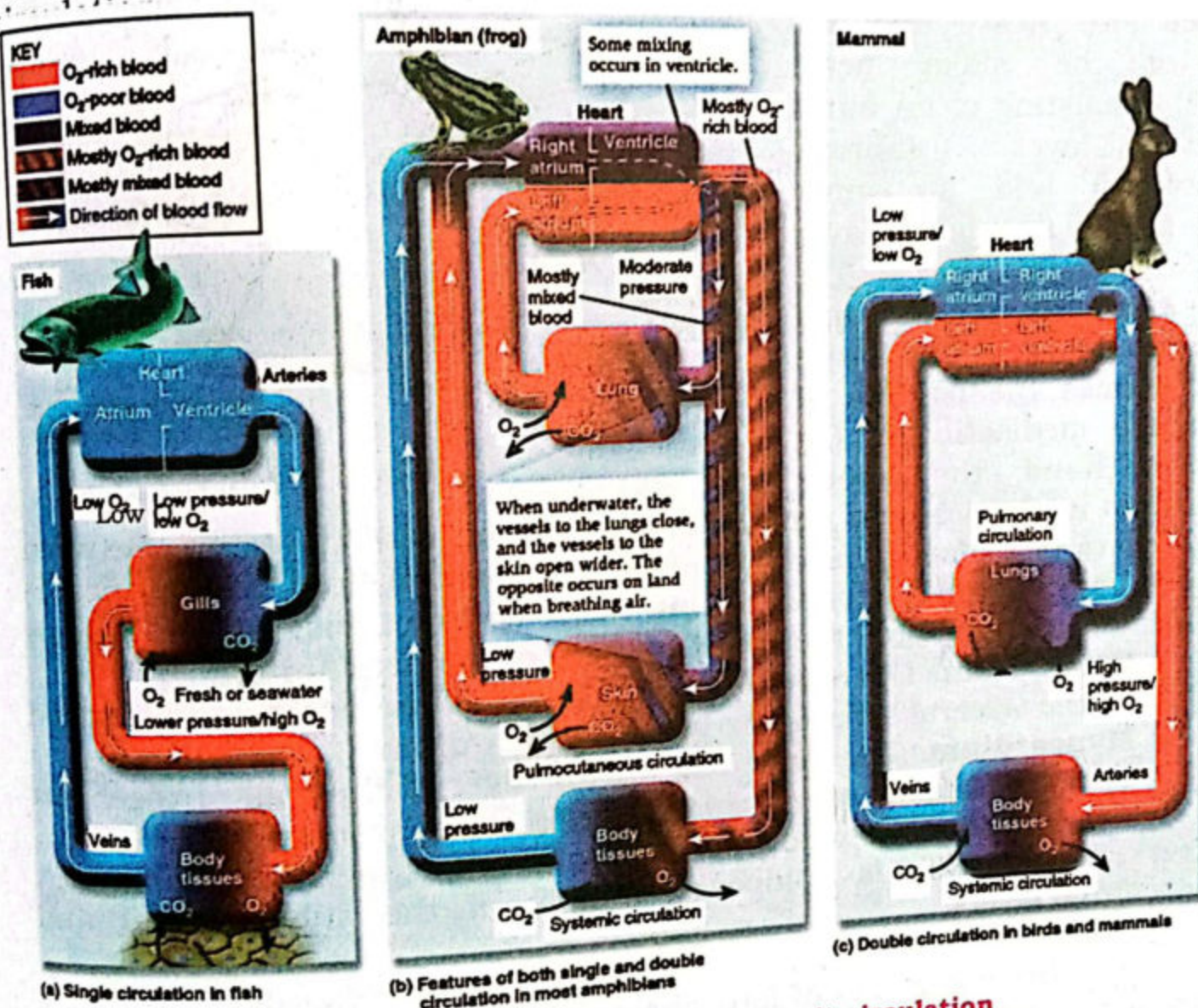


Fig: 12.2 Single and double circuit circulation

12.2 GENERAL CHARACTERISTICS OF A CIRCULATORY SYSTEM

- All circulatory systems have three major components.
- A fluid, **blood** that serves as a medium of transport.
 - A system of channels or **vessels** that conduct the blood throughout body.
 - A pump, the **heart** or modified blood vessel that keeps the blood circulating.

12.3 HUMAN HEART

The human heart has a somewhat conical form and is enclosed by pericardium. It is located between lungs with one third situated on the right and two thirds on the left of the midline, just behind sternum.

Layers of heart wall

The heart wall consists of three layers enclosed in the pericardium. The pericardium is the thick covering that encloses the heart and the roots of the major heart vessels, consisting of an outer fibrous layer (fibrous pericardium) and an inner double serous membrane layer (serous pericardium). The fibrous pericardium consists of thick fibrous connective tissue and it defines the borders of the middle mediastinum. On the other hand, the serous

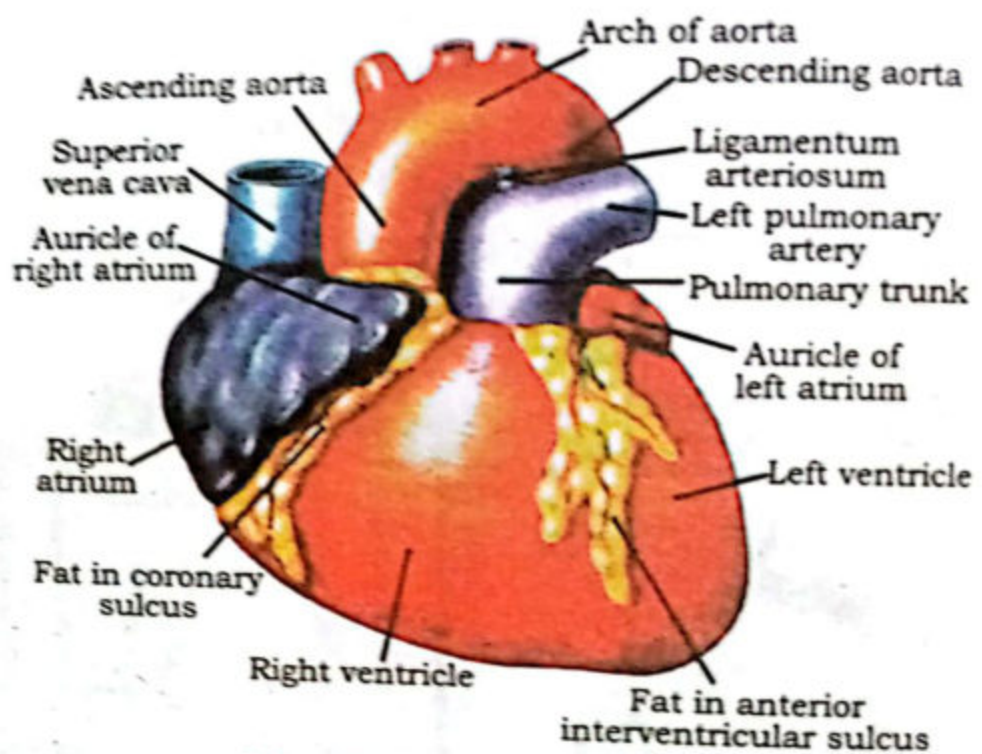


Fig: 12.3 Human heart

pericardium is physically in a much closer relation with the heart. Between its two layers is a small amount of serous pericardial fluid that lubricates the layers and prevents friction during heart contractions, which is along with mechanical protection, the basic function of the pericardium.

- I. **Epicardium.** The outer layer of the wall of the heart and is formed by the visceral layer of the serous pericardium.
- II. **Myocardium.** The muscular middle layer of the wall of the heart and has excitable tissue and the conducting system.
- III. **Endocardium.** A middle concentric layer, a subendocardial layer. The rest of the heart is composed mainly of the subepicardial and subendocardial layers.

12.3.1 Structure of heart

Human heart consists of four chambers, two upper thin walled atria and two lower, thick walled ventricles. Two large veins, superior and inferior vena cava enter the right atrium and two pairs of pulmonary veins open into the left atrium. Similarly, two large arteries emerge out, one from the right ventricle, the **pulmonary aorta** and other from the left ventricle the **systemic aorta**.

The right and left atria are separated by a vertical membranous inter atrial septum, the right atrium and left ventricles are also separated by thick muscular interventricular septum. Four valves in the heart each consisting



of flaps of connective tissue, prevent backward flow of blood. The right atrium opens into the right ventricle by an artioventricular valve (Tricuspid valve). The left atrium opens into left ventricle by artioventricular valve (Bicuspid valve or mitral valve). The **AV valves** are anchored by strong fibers that prevent them from turning inside out. Pressure generated by the powerful contraction of the ventricles closes the AV valves keeping blood from flowing back into the atria. Semilunar valves are located at the two exits of the heart, where the systemic aorta leaves the left ventricle and the pulmonary aorta leaves the right ventricle. The blood is pumped out into the arteries through the semilunar valves, which are forced open by pressure created by ventricular contraction when the ventricles relax blood starts to flow back towards the heart, closing the semilunar valves, which prevents blood from flowing back into the ventricles. The cavity of the left ventricle is narrower than the right ventricle because of more muscular walls. It is due to the fact that the right ventricle has to pump blood into the lungs only (pulmonary circulation) while the left ventricle pumps blood to the entire body (systemic circulation).

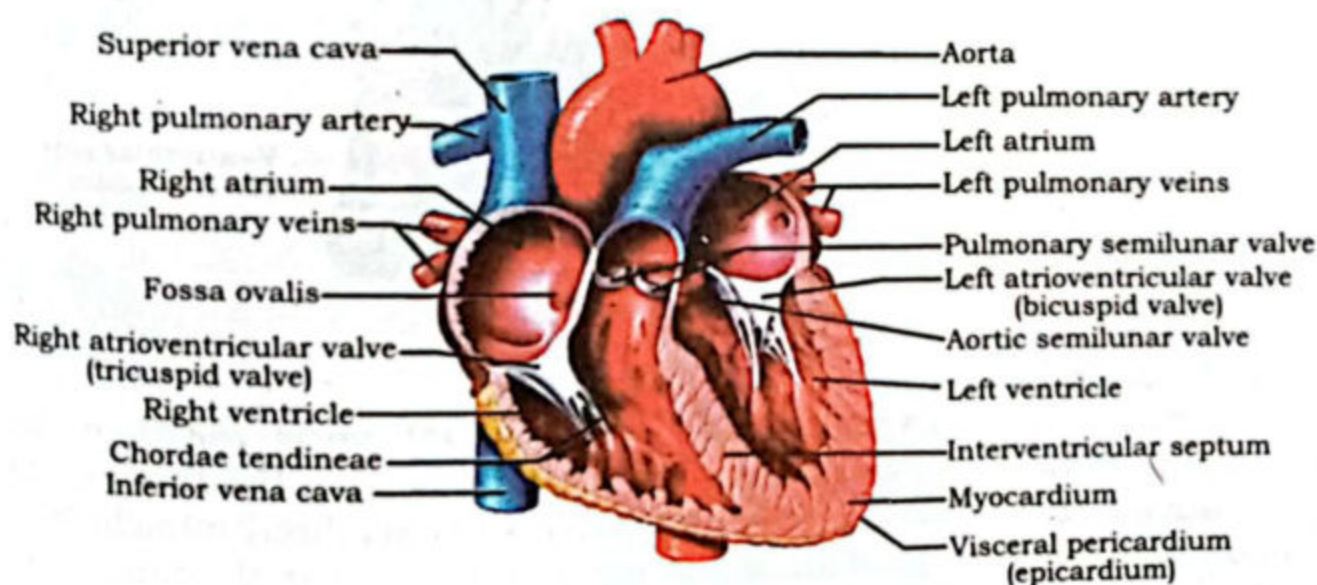


Fig: 12.4 Anatomy of heart

12.3.2 Phases of heart beat

The contraction of the heart chambers are known as **heart beats**, which are rhythmically and regular. A human normal **heart beats** 72 times per minute at rest. The alternating contraction and relaxation of the heart chambers is called the **cardiac cycle**. The two atria contract synchrony, emptying their contents into the ventricles. A fraction of a second later, the two ventricles contract simultaneously forcing blood into aorta leaving the heart.

Both chambers than relax briefly before the cycle-repeated. The period of ventricular contraction is called **ventricular systole**. The rest of

the cycle, including relaxation of all the chambers followed by contraction of the atria, is called **ventricular diastole**. At normal resting heart rate systole lasts about 0.3 second and diastole about 0.5 second. When the heart beats the familiar **LUB-DUB** sound is heard as the valves of the heart close, the **LUB** is caused by vibrations of heart when the artioventricular valves close, and **DUB** is heard when vibration occurs due to the closing of the semilunar valves.

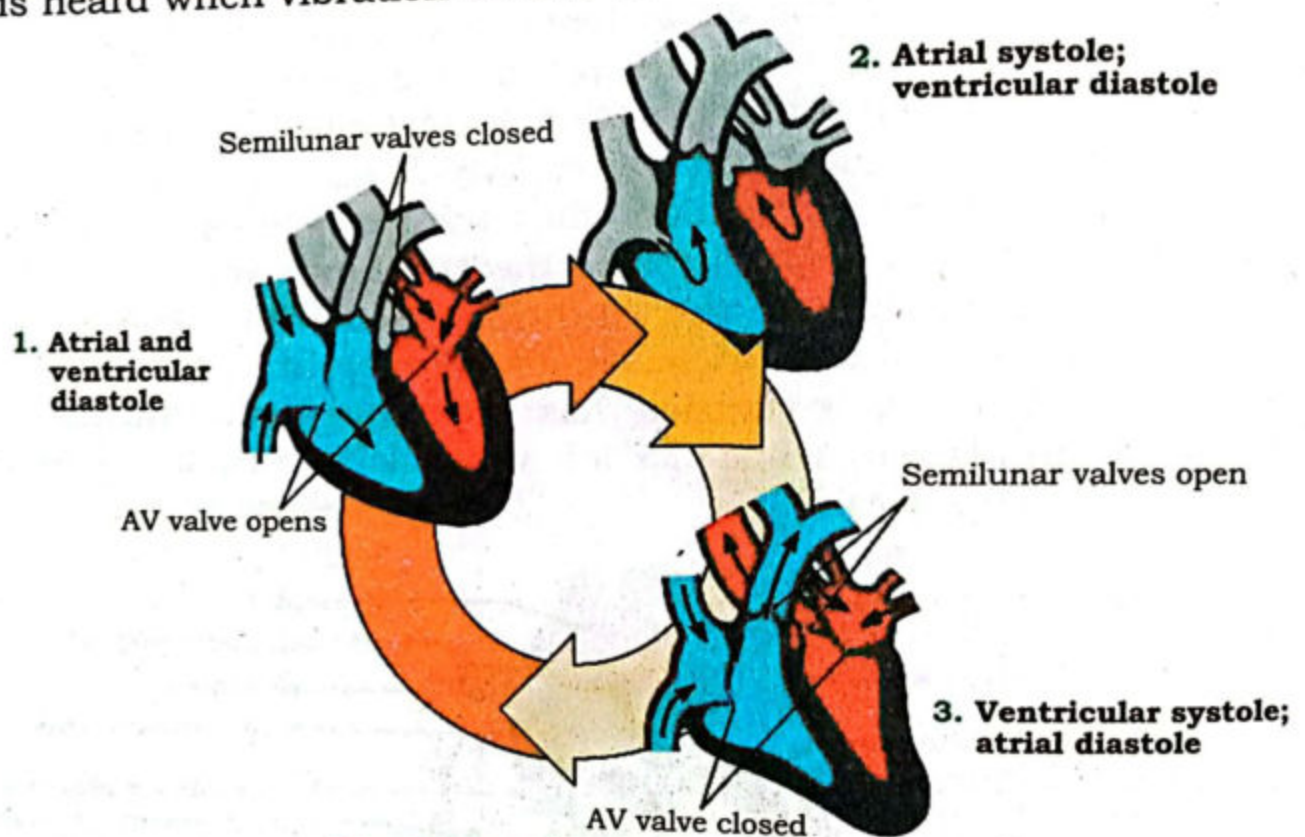


Fig: 12.5 Cardiac cycle

12.3.3 S.A Node and A.V Node

A region of the heart called the **saino atrial node (SAN)**, or **pace maker**, maintains the heart's pumping rhythms by setting the rate at which all cardiac muscle cells contract. It is composed of specialized muscle tissue. The SA node is located in the wall of the right atrium, near the point where the superior vena cava enters the heart. The SA node generates electrical impulses much like those produced by nerve cells. Impulses from the SA node spread rapidly through the walls of the atria, making them contract in unison. The impulses also pass to another region of specialized muscle tissue, a relay point called the atrioventricular (AV) node in the wall between the right atrium and right ventricle. Here the impulses are delayed for about 0.1 second which ensures that the atria will contract first and empty the ventricles, through two bundles of specialized muscle fibers of atrioventricular bundle or **bundle of His** in the ventricular septum and then into the wall of the ventricle through a network of fibers called **purkinji fibers**.



12.4 ELECTROCARDIOGRAM (ECG)

The impulses that travel through cardiac muscle during the heart cycle produce electrical currents that are conducted through body fluids to the body surface where the currents can be detected by electrodes placed on the skin and recorded as an electrocardiogram (ECG)

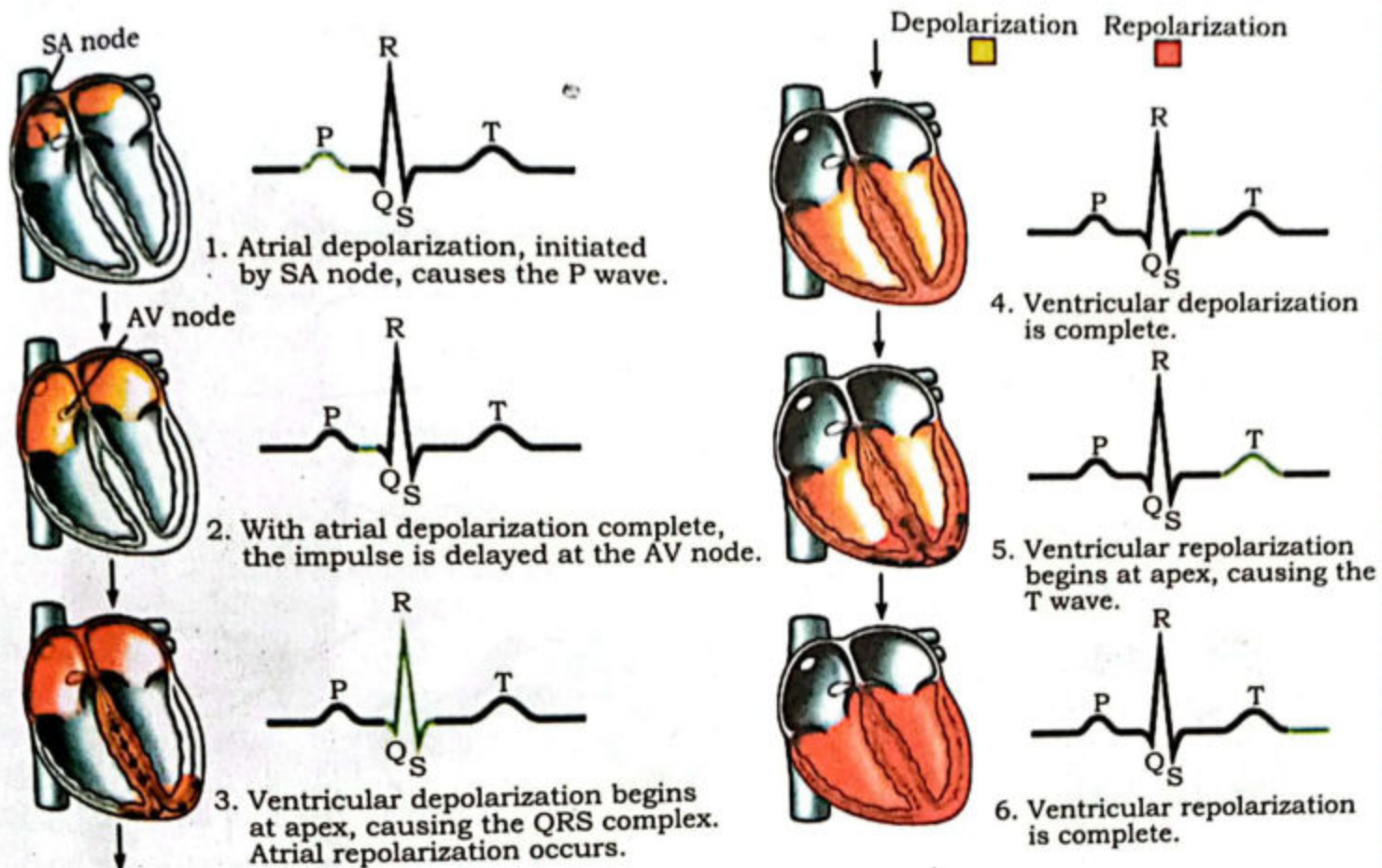


Fig: 12.6 Electrocardiograms

12.4.1 Uses of ECG

By the help of electrocardiogram, we can see the electrical changes in the form of depolarization and repolarization. Depolarization of the atrial fibers of the SA node produces the **P wave**. The ventricles of the heart are in diastole during the expression of the P wave on the ECG recording. The **P-R interval** is the period of time start of the P wave to the beginning of the QRS complex this interval indicates the amount of time required for the SA depolarization to reach the ventricles. The **QRS complex** begins as a short downward deflection (**Q**). Continue as a sharp upward spike (**R**) and end as downward deflection (**S**). The QRS complex indicates the depolarization of the ventricles during this interval, the ventricles are in systole and blood is being ejected from the heart. The time duration known as the **S-T segment** represents the period between the completion of ventricular depolarization and initiation of repolarization. The **T wave** is produced by ventricular repolarization. A normal ECG indicates that the heart is functioning

properly. The second wave, or the QRS complex occurs just prior to ventricular contraction the third, or T wave occur just before the ventricles relax.

12.5 BLOOD VESSELS

As blood leaves the heart, it travels from **Aorta** to arteries to arterioles to capillaries to venules to vein, and then **vena cava** which return it finally back to the heart. In this section, we examine each type of blood vessel in more detail.

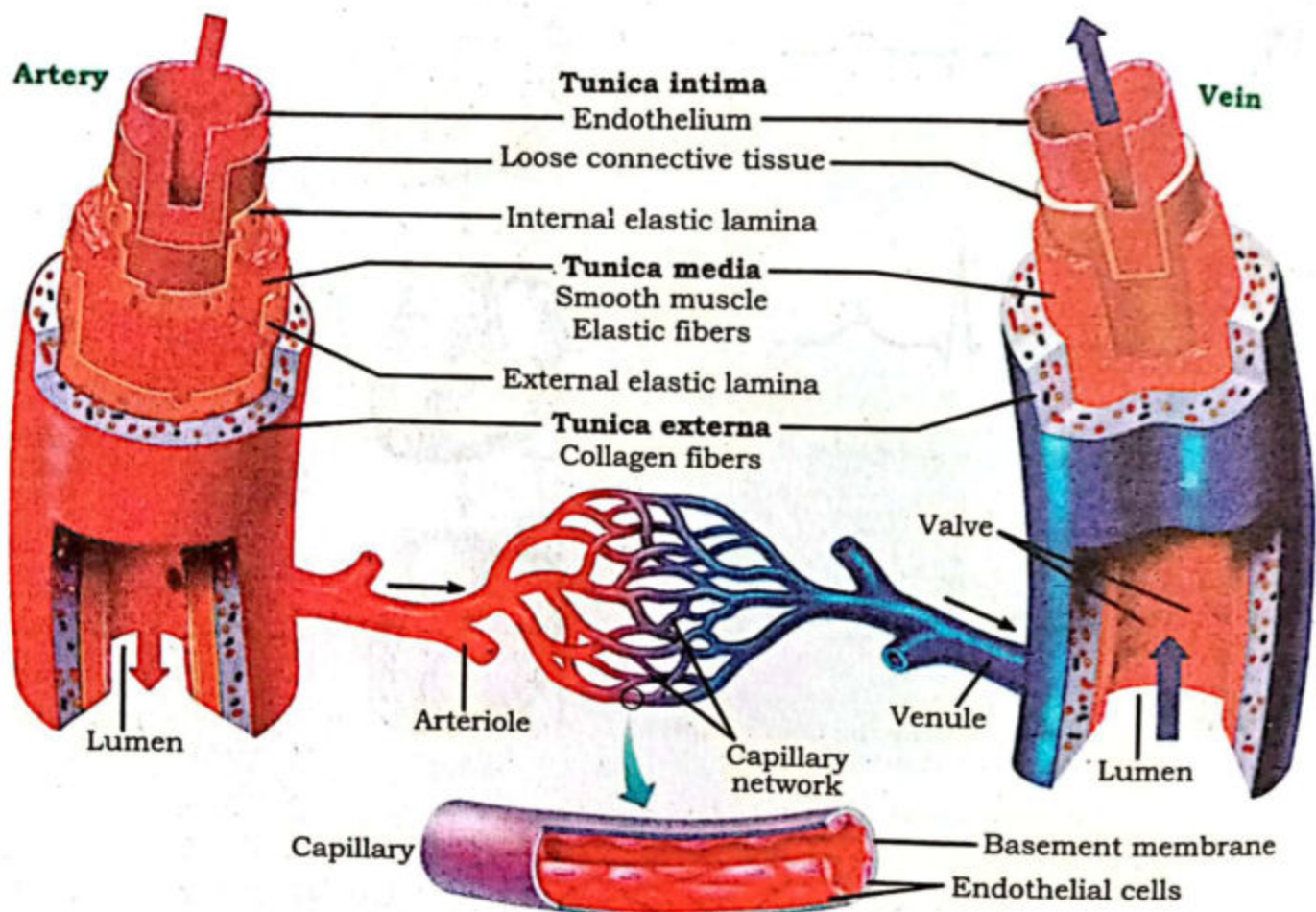


Fig: 12.7 Blood vessels

12.5.1 Arteries

After leaving the heart blood first enters large vessels called arteries. These are thick-walled vessels consisting of three layers. The outer **tunica externa** composed of fibrous connective tissue having collagen fibers. The middle layer **tunica media** has smooth muscles and elastic fibers. The inner **tunica interna** (intima) consists of squamous endothelium. The arteries expand slightly, like thick-walled balloons. Between heart beats they recoil, helping to pump the blood and maintain a steady flow through the smaller vessels. Arteries branch into vessels of even smaller diameter called arterioles, which play a major role in determining how blood is distributed within the body.



12.5.2 Capillaries

The entire circulatory system is an elaborate device for providing each cell of a complex, multicellular organism with the type of exchange-diffusion-practiced by the simplest unicellular organisms. This is accomplished at the level of the capillaries, the tiniest of all vessels. Here wastes, nutrients, gases, and hormones are exchanged between the blood and the body cell. Their walls consist of a single layer of endothelium, which presents very little resistance to the diffusion of dissolved substances in or out. The capillaries are extremely narrow (7-9 μ in diameter). The pressure within capillaries causes a continuous leakage of fluid from the blood plasma into the spaces surrounding the capillaries and tissues. This fluid is known as the **interstitial fluid** and consists primarily of water in which are dissolved nutrients, hormones, gases, wastes and small proteins from the blood. The large plasma proteins, red blood cells, and platelets are unable to leave the capillaries because of their size, although white blood cells can ooze through the openings between capillary cells.

12.5.3 Veins

Blood from the capillaries, now carrying carbon dioxide and other cellular waste drains into larger vessels called venules that empty into still larger veins. Veins provide a low resistance pathway by which blood can return to the heart. The walls of veins are much thinner and more expandable than those of arteries although both contain a layer of smooth muscles because blood pressure in the veins is low, the contractions of skeletal muscle during exercise and breathing must assist in the return of blood to the heart. These muscular movement squeeze the veins, forcing blood through them. When veins are compressed, you might predict that blood would be forced away from the heart as well as toward it. To prevent this, veins are equipped with one-way valves that allow blood flow only toward the heart.

12.6 ROLE OF ARTERIOLES IN VASODILATION AND VASOCONSTRICTION

The muscular walls of arterioles are under the influence of nerves, hormones and chemicals produced by nearby tissues. They can therefore contract and relax in response to the changing needs of the tissues and organs they supply. On the hot summer day, however you become flushed as the arterioles in your skin expand (vasodilation) and bring more blood to the skin capillaries to dissipate excess heat to the outside and to maintain a proper internal temperature. In contrast in extremely cold weather, finger and toes can become frostbitten because the arterioles supplying the extremities constrict.

12.7 ROLE OF PRE-CAPILLARY SPHINCTER IN REGULATING THE FLOW OF BLOOD THROUGH CAPILLARIES

The flow of blood in capillaries is regulated by tiny rings of smooth muscles pre capillary sphincters. That surrounds the junctions between arterioles and capillaries. These sphincters open and close in response to local changes that signal the needs of nearby tissues for examples, accumulation of carbon dioxide, lactic acid or

other cellular wastes signals the need for increased blood flow to the tissues. These signals cause pre capillary sphincters as well as the muscles in nearby arterioles to relax, thus increasing blood flow through the capillaries.

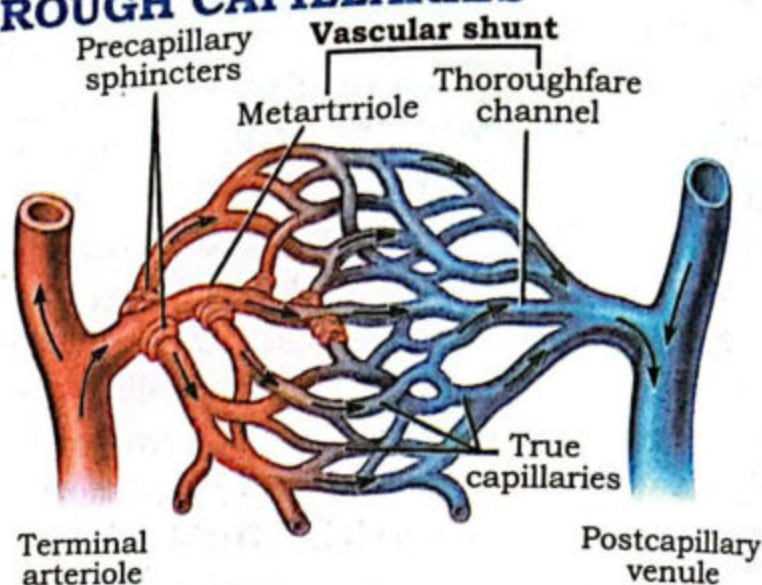


Fig: 12.8 Capillary network

12.8 VASCULAR PATHWAY

The human cardiovascular system includes two major pathways. Pulmonary circulation and systemic circulation.

12.8.1 Pulmonary circulation

In the pulmonary circulation, pathway of blood can be traced as follows. Deoxygenated blood from whole body collects in the right atrium and then passes into the right ventricle which pumps deoxygenated blood into the pulmonary aorta. The pulmonary aorta divides into right and left pulmonary arteries, which carry blood to lungs as blood passes through the pulmonary capillaries, carbon dioxide is given off and oxygen is picked up. Blood returns to the left atrium of the heart through pulmonary veins.

12.8.2 Systemic circulation

The systemic circulation includes aorta and vena cava as the major pathways for blood. In the systemic circulation arteries carry oxygenated blood and have a bright red colour, but veins carry deoxygenated blood and appear dull red or, when viewed through skin, blue.

12.8.3 Coronary circulation

The coronary arteries supply blood to wall of the heart (myocardium). The coronary arteries arise from the aorta just above the aortic semilunar valve. They lie at the exterior surface of the heart, where they branch into arterioles and the capillaries. The capillary beds enter venules which join to form cardiac veins, and these empty into right atrium.

12.8.4 Hepatic portal system

A portal system is one that begins and ends in capillaries. One place in human body where a portal system is found is between small intestine



and the liver. Blood passes from the intestinal villi into venules that join to form the hepatic portal vein, a blood vessel that connects the intestine with the liver. The hepatic vein leaves the liver and enters the inferior vena cava.

12.8.5 Renal circulation

The renal arteries are short and spring directly from abdominal aorta, enters into kidney and gives branches which pass through medulla. In cortex they give rise to glomerular arterioles from here blood enter in the peritubular capillaries and vasa recta, from these capillary networks the blood is drained through veins and leave the kidney as a single renal vein that empties into the inferior vena cava.

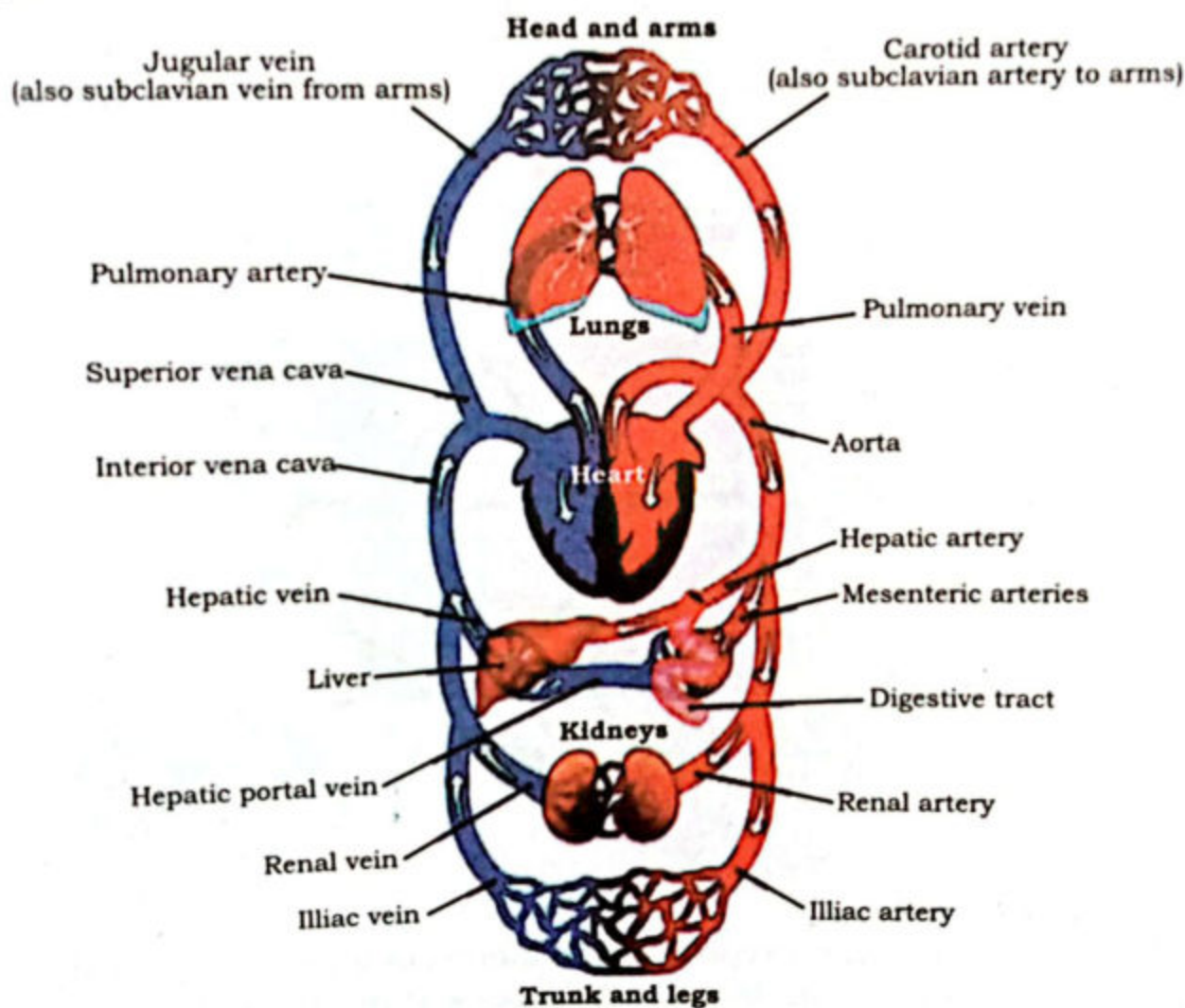


Fig: 12.9 cardiovascular system

12.9 RATE OF BLOOD FLOW IN BLOOD VESSELS

Blood flows through the vessels at an uneven speed. It flows faster in large arteries and much slower in capillaries network. Although an individual capillary is much narrower but the capillary beds have an enormous number of such capillaries, so that the total diameter of these vessels is much greater than the arteries. For this reason, the blood flows slowly in the capillaries, permitting the exchange of materials between the

blood and interstitial fluid. As blood leaves the capillary bed and passes to the venule and veins it speeds up again due to the reduction of in total cross-sectional area.

12.10 BLOOD PRESSURE

The hydrostatic force of blood exerted on the wall of vessels is called blood pressure. Blood pressure is the main force that propels blood from the heart through the arteries, arterioles to the capillary beds. Fluid always flows from area of higher pressure to area of low pressure. Blood pressure measured with sphygmomanometer. It is normally measured on the brachial artery, which is in the upper arm, and is stated in millimeters of mercury (mmHg). Blood pressure readings consist of two numbers 120/80 mmHg. First the systolic pressure (120 mmHg) and second diastolic blood pressure (80 mmHg).

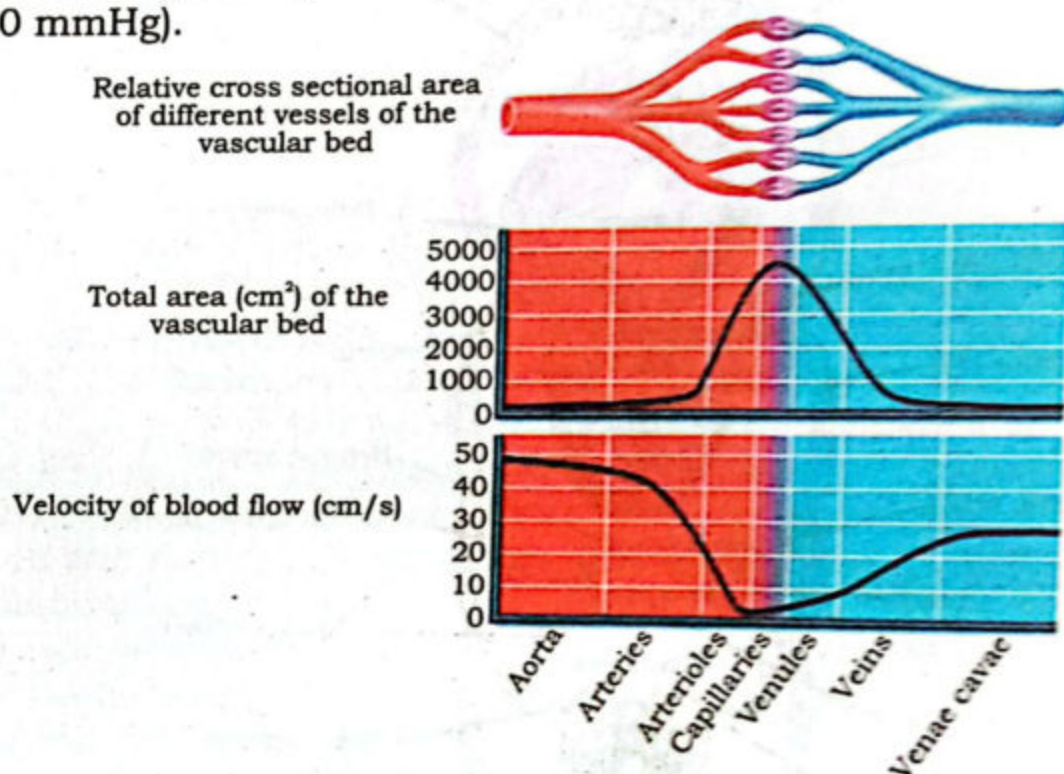


Fig: 12.10 Velocity of blood flow through blood vessels

12.10.1 Baroreceptors

Baroreceptors are a type of mechanoreceptors allowing for the relay of information derived from blood pressure within the autonomic nervous system. There are two types of baroreceptors:

High-pressure arterial baroreceptors

They sense the blood pressure and relay the information to the nervous system, so that a proper blood pressure can be maintained.

Low pressure volume baroreceptors

These are found in the large veins and in the walls of the atria of the heart. The low pressure baroreceptors are involved with the regulation of blood volume.



12.11 CARDIOVASCULAR DISORDERS

Diseases of heart, blood vessels and blood circulation are generally termed as cardiovascular disorders.

12.11.1 Thrombus

The formation of blood clot (thrombus) within an intact blood vessel is initiated by atherosclerotic plaques. The plaques when destroy the endothelium of the blood vessels, platelets gather at the damaged site to initiate the process of clot. **Thrombus** and **embolus** are potentially life threatening blood vessel problems that increase the risk of having a heart attack or stroke. Thrombus happens when a blood clot or thrombus grows in blood vessels. This can reduce blood flow. An embolus is any foreign material that travels within the body. If it becomes stuck and severely blocks the flow of blood. The issue is called embolism.

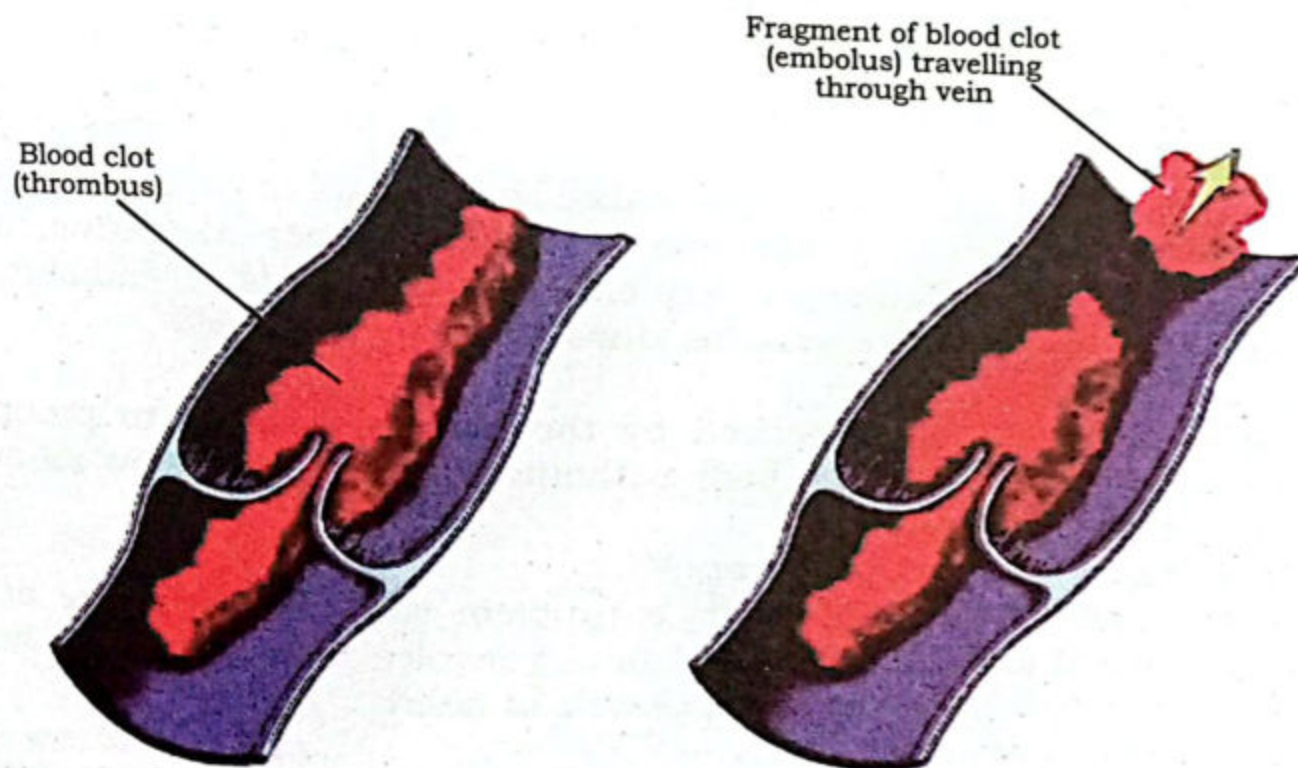


Fig: 12.11 Thrombus and embolus

12.11.2 Atherosclerosis

It is a disease of the arterial wall (intima) which loses its elasticity gradually its inner layer thickens causing narrowing of the artery and consequently impairing the blood flow. The narrowing is due to the formation of fatty lesions called atheromatous plaques in the inner lining of the arteries. These plaques consist of low density lipoproteins (LDL or cholesterol and protein) decaying muscle cells, fibrous tissue, clumps of blood platelets and sometimes calcium. The arteries become extremely hard and the disease is called arteriosclerosis or simply hardening of the arteries.



Factor causing atherosclerosis

The possible causes of atherosclerosis are smoking, hypertension, male gender, obesity physical inactivity, a high serum cholesterol level, severe diabetes, family history of arterial disease and possibly an anxious or aggressive personality. The risk of atherosclerosis increases with age.

12.11.3 Heart problems

Cardiovascular disorders related to heart called heart problems. It includes angina pectoris, heart attack and heart failure.

Angina pectoris

Strain on the heart by increasing resistance to blood flow. Although the heart may enlarge in response to this added demand, its own blood supply may not increased proportionately. The heart muscle is then inadequately supplied with blood, especially during exercise. Lack of sufficient oxygen to the heart can cause chest pain and a radiating pain in the left arm called angina pectoris.

Heart attack

A myocardial infarction also called heart attack, occurs when a portion of the heart muscle dies due to lack of oxygen it is due to the blockage of any of the coronary artery either by thrombus or embolus, the blood supply to some cardiac muscles stops.

Heart failure

Heart failure is characterized by the heart's inability to pump an adequate supply of blood to the body without sufficient blood flow all major body functions are disrupted.

12.11.4. Congenital heart problem

A congenital heart problem is a problem with the structure of the heart. It is present at birth. The problem can involve the walls of the heart, the valves of the heart, and the blood vessels of heart.

12.11.5 Principles of angiography

Angiography is the x-ray imaging of blood flow in the body. During an angiogram, substances that are opaque to x-rays are inserted into the bloodstream. The images of the path they take through blood vessels can be diagnostically useful. It is useful to locate blockages in the blood vessels.

12.11.6 Treatment of cardiovascular disorders

Coronary bypass

It is a medical procedure to improve blood flow to the heart. It may be needed when the arteries supplying blood to heart tissue, called coronary arteries are narrowed or blocked. Coronary bypass uses blood vessels from another part of the body and connects them to blood vessels above and below the narrowed artery, bypassing the narrowed or blocked coronary arteries.

Angioplasty

It is a medical procedure used to improve the flow of blood through narrowed artery. It involves placing a long thin tube (catheter) into a blood vessel to place a small balloon at the site of the narrowing. When the balloon is inflated, the blood vessel should open and allow a healthy flow of blood.

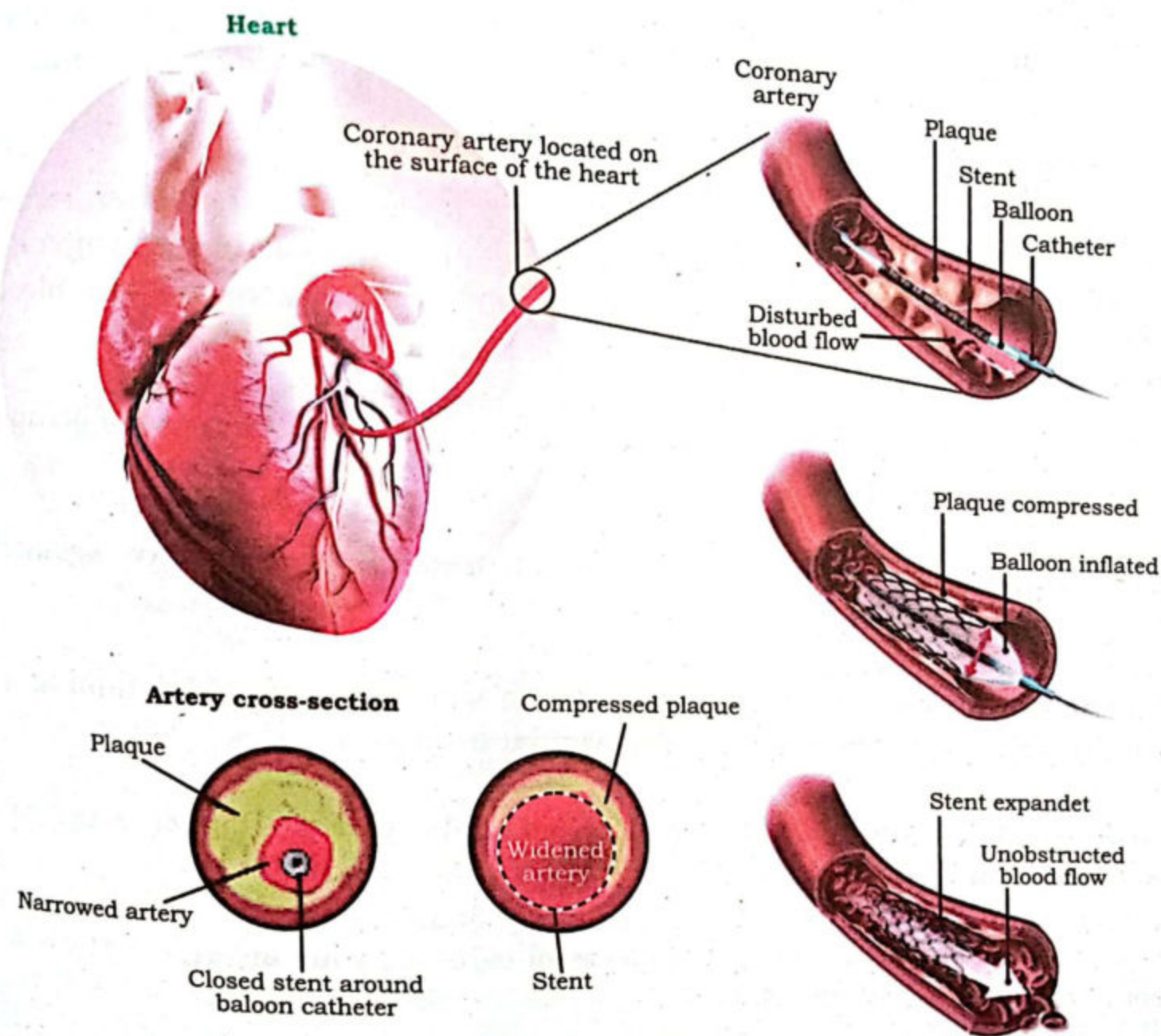


Fig: 12.12 Balloon angioplasty

Open heart surgery

It is a surgical procedure in which breast-bone being cracked open, providing direct access to the heart, allowing surgeons to fix heart problems while the patient is placed on a heart-lung machine. Open heart surgery is probably the most widely known and feared of all operations, but in the right hands open heart surgery can be remarkably safe and effective, associated with excellent outcomes for patients.



12.11.7 Hypertension

When the mean arterial pressure is greater than the upper range, women of any age are considered to have hypertension if their blood pressure reading is 160/95 or above. For a man under age 45, a reading above 140/95 is considered hypertensive. Hypertension is called as "Silent Killer", because the affected individuals may show no outward symptoms until a stroke or heart attack occurs. Several factors such as heredity, higher intake of salts in diet, smoking, obesity and disorders of kidney or adrenal gland are responsible for hypertension.

Hypotension (low blood pressure)

When the mean arterial pressure is lower than the lower range, people with low blood pressure (Hypotension) may experience symptoms when their blood pressure drops below 90/60. Symptoms of low blood pressure fatigue, dizziness, nausea and lightheadedness.

12.11.8 Life style modification prevents cardiovascular disorders

There are several things you can do to reduce your chances of getting cardiovascular disorders.

Eat healthy diet:

Low salt and low saturated fats and cholesterol as protective against cardiovascular disorders.

Stay at a healthy weight:

Hypertension occurs more often in persons who are obese. Maintaining a healthy weight reduces risk of cardiovascular disorders

Don't smoke:

Smoking raises your blood pressure so puts you at higher risk for cardiovascular disorders

Keep your cholesterol and triglyceride levels under control:

High level of saturated fats and cholesterol can clog your arteries and raise risk of cardiovascular disorders.

Exercise:

Those who exercise are less apt to have cardiovascular disorders.

12.12 LYMPHATIC SYSTEM

The lymphatic system consists of a network of blind vessels (lymphatics) that drains lymph from all over the body back to the circulatory system. Numerous small lymph nodes, and two additional organs thymus and spleen although not strictly part of the circulatory system, the lymphatic system is closely associated with it.



12.12.1 Composition and formation of intercellular fluid

All the body tissues are bathed in a watery fluid derived from the blood stream. This intercellular or tissue fluid is formed when blood passes through the capillaries. The capillary walls are permeable to all components of blood except the R.B.Cs and blood proteins.

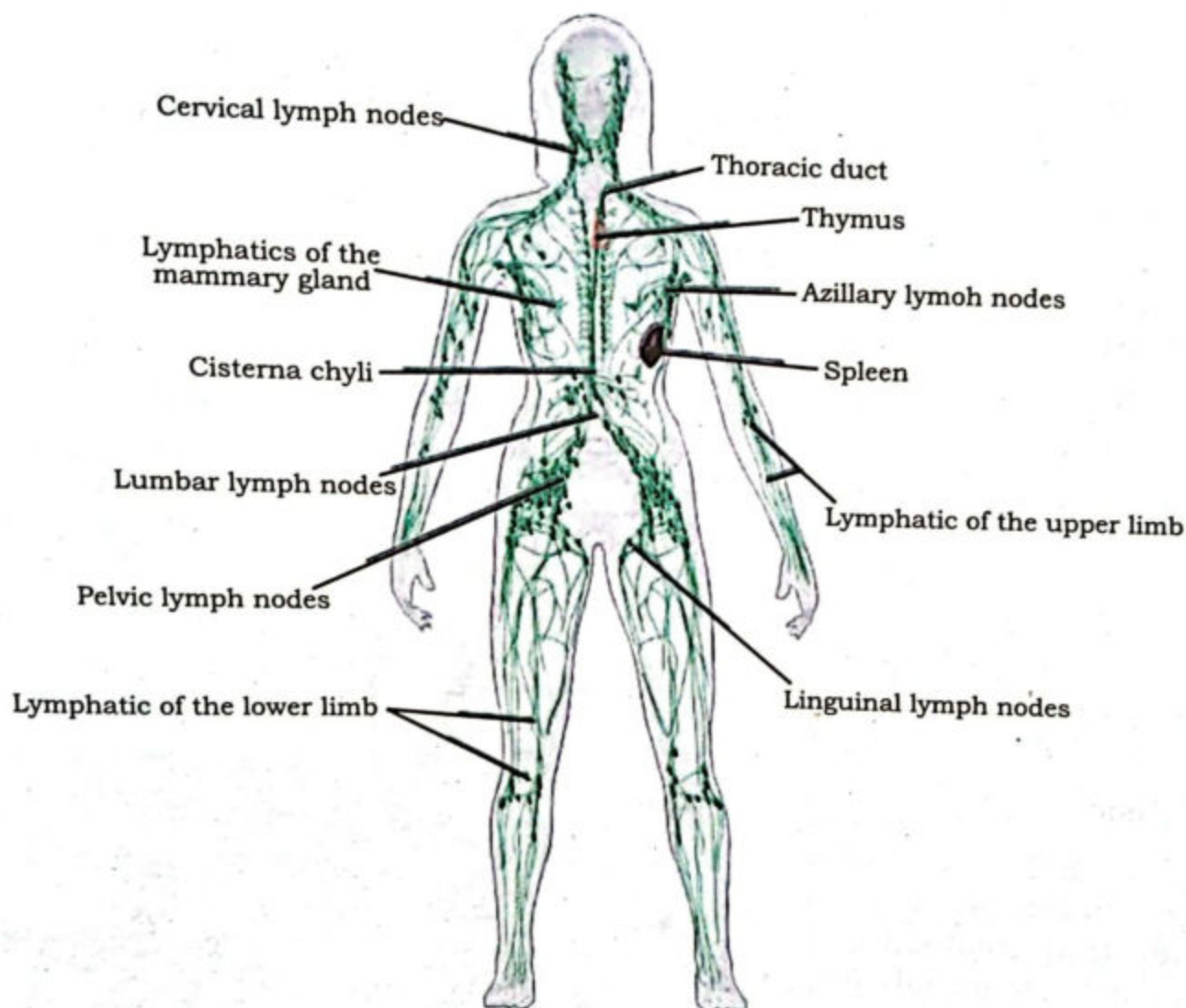


Fig: 12.13 Lymphatic system

12.12.2 Comparison of the composition of intercellular fluid and lymph

The fluid passes from the capillary into the intercellular spaces as the intercellular or tissue fluid. About 85% fluid returns into the blood at the venous end of the capillary. The rest 15% of the tissue fluid drains into the blindly ending lymphatic capillaries as lymph along with W.B.Cs, cell debris and microorganisms like bacteria, are transported back to the heart through lymphatic system. Thus lymph can be defined as colourless body fluid that contains lymphocytes, small proteins and fats. Lymph takes fluid substances from cells of tissues and intercellular spaces, which cannot penetrate the blood capillaries.

12.12.3 Lymphatic vessels

Like blood capillaries, **lymph capillaries** form a complex network of microscopically narrow, thin walled vessels into which substances can move readily. Lymph capillary walls are composed of cells with openings between them that act as one way valves. These openings allow relatively large particles along with fluid, to be carried into lymph capillary. Materials collected by the lymph capillaries flow into larger **lymph vessels**, through which lymph is forced as blood is through veins. Large vessels have somewhat muscular walls, but most of the impetus for lymph flow comes from the contraction of nearby muscles, such as those used in breathing and walking. As in blood veins, the direction of flow is regulated by one-way valves. Efferent lymph vessels merge before entering one of two ducts: the thoracic trunk or the right lymphatic trunk

12.12.4 Role of lymph vessels in villi

The small intestine is richly supplied with lymph capillaries. After absorbing digested fats, intestinal cells release fat globules into the interstitial fluid. These globules are too large to diffuse into blood capillaries but can move easily through the openings between lymph capillary cells. These fat globules may make up 1% of the lymphatic fluid.

12.12.5 Lymphoid tissue

The large lymph vessels are interrupted periodically by kidney bean shaped structures about one inch long called **lymph nodes**. Lymph is forced through channels with in the nodes that are lined with masses of macrophages.

Lymphocytes are also produced in the nodes. Both of these white blood cells recognize and destroy foreign particles, such as bacteria and viruses, and are killed in the process.

Spleen

The spleen is located in the left side of the abdominal cavity, between the stomach and diaphragm. The spleen filters blood, exposing it to macrophages and lymphocytes that destroy foreign particles and aged blood cells.

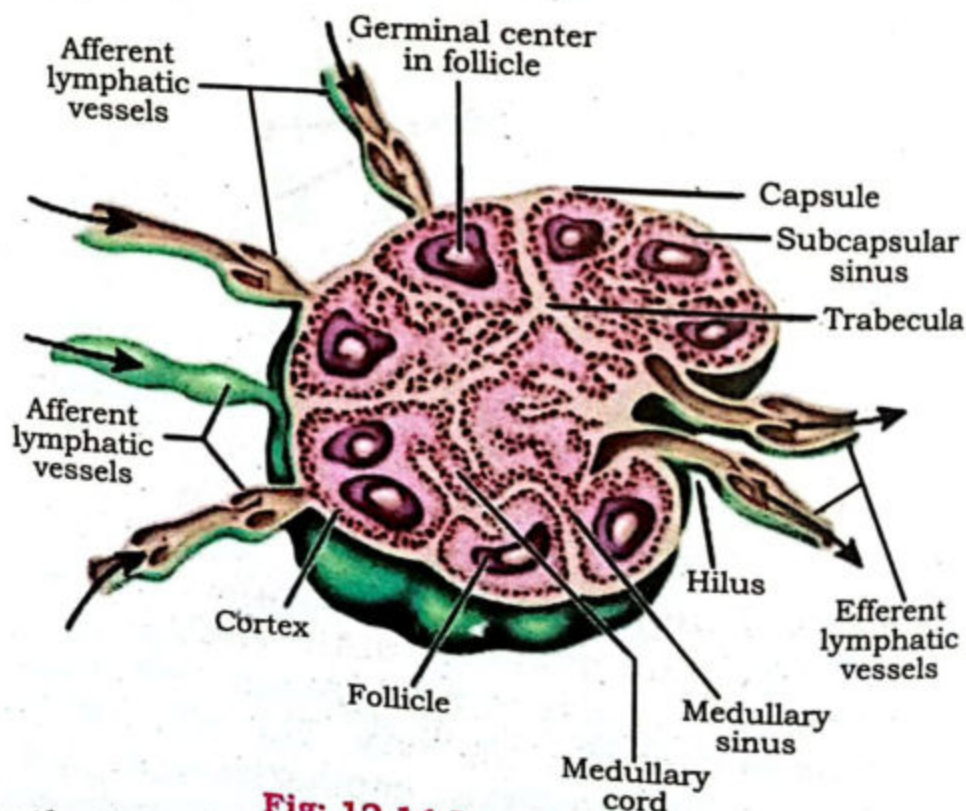


Fig: 12.14 Lymph node



SUMMARY

- Living organisms obtain necessary raw materials, to synthesize molecules for metabolism. These materials are transported with the living organisms or from environment to living organism.
- The circulatory system is of two types: open type and closed type.
- In arthropods and molluscs blood bathes the internal organs directly.
- There is no distinction between blood and interstitial fluid, and the general body fluid is more correctly termed as haemolymph.
- In a close type circulatory system, blood is confined to vessels and is distinct from the interstitial fluid.
- The heart of fish is two chambered consisting of an atrium and a ventricle.
- From amphibians onwards up to the mammals, the circulatory system has double circuit plane with separate pulmonary and systemic circulation.
- In amphibians and reptiles, the heart consists of three chambers, two atria and one ventricle.
- In crocodiles, birds and mammals the heart is four chambered with two atria and two ventricles.
- The right atrium and right ventricle are completely separated from the left atrium and left ventricle by inter atrial and inter ventricular septum.
- The human heart has somewhat conical form and is enclosed by pericardium.
- The heart wall consists of three layers enclosed in the pericardium.
- Human heart consists of four chambers. Two upper thin walled atria and two lower thick walled ventricles.
- The contractions of heart chamber are known as heartbeat.
- A region of the heart called the sino atrial (SA) node or pace maker, maintains the heart's pumping rhythms.
- As blood leaves the heart, it travels from arteries to arterioles to capillaries to venules to vein, which return it finally to the heart.
- The hydrostatic force of blood exerted on the wall of vessels is called blood pressure.

EXERCISE

1. Encircle the correct choice.

- (i) Lymph most closely resembles which of the following?
- | | |
|------------|------------------------|
| (a) Blood | (b) Urine |
| (c) Plasma | (d) Interstitial fluid |
- (ii) What event initiates blood clotting?
- (a) Contact with an irregular surface by platelets and other factors in plasma.
- (b) Production of the enzyme thrombin
- (c) Conversion of fibrinogen into fibrin
- (d) Excess flow of blood through a capillary
- (iii) Damage to the S.A Node in humans
- (a) Is a major contributor to heart attacks.
- (b) Would block conductance between the bundle branches and the purkinje fibers.
- (c) Would have a negative effect on peripheral resistance
- (d) Would disrupt the rate and timing of cardiac muscle contractions.
- (iv) Which of the following is measured by an electrocardiogram?
- (a) Impulses from the AV node
- (b) Impulses of the parasympathetic nervous system that control heartbeat
- (c) The speed on impulses from the SA node
- (d) Contraction of the two atria
- (v) Where are semilunar valves to be found in the mammalian heart?
- (a) Where blood goes from atria to ventricles
- (b) On the right side of the heart only
- (c) Where the pulmonary veins attach to the heart
- (d) At the places where blood leaves via the aorta and pulmonary arteries
- (vi) Organisms in which a circulating body fluid is distinct from the fluid that directly surrounds the body's cells are likely to have which of the following?
- (a) An open circulatory system
- (b) A closed circulatory system
- (c) A gastrovascular cavity
- (d) Branched tracheae
- (vii) Average blood pressure is lowest in which structure (s)
- | | |
|-----------------|----------------|
| (a) Aorta | (b) Arterioles |
| (c) Capillaries | (d) Vena cava |

- (viii) Which of the following is/ are a cause (s) of vasoconstriction?
- (a) Lying down after standing
 - (b) Standing after lying down
 - (c) Stress or hormone concentration
 - (d) Histamine secretion
- (ix) Blood returning to the human heart in a pulmonary vein drains first into the?
- (a) Left atrium
 - (b) Right atrium
 - (c) Left ventricle
 - (d) Right ventricle
- (x) Which of the following is NOT an important function of the human circulation?
- (a) Transport of nutrients and respiratory gases
 - (b) Regulation of body temperature
 - (c) Protection of the body by circulating antibodies
 - (d) Removal of waste products for excretion from the body

2. Write short answers of the following questions:

1. Why higher animals need circulatory system?
2. Why SA node is called pace maker of heart?
3. Why the capillaries have a single layer of endothelium?
4. What are LUB and DUB?
5. Why circulatory system of arthropods is called open type?
6. Why blood flow faster in arteries?
7. Distinguish between single circuit circulation and double circuit circulation.

3. Write detailed answers of the following questions:

1. Describe the structure of blood vessels in man.
2. Describe the structure of human heart.
3. What is circulation? Explain needs of circulation
4. State the phases of heartbeat in man.
5. Describe the flow of blood through human heart as regulated by the valves.
6. Describe the lymphatic system of man
7. Explain the electrocardiogram with the help of diagram.
8. Describe pulmonary circulation and systemic circulation